MRes Neuroergonomics Data Challenge

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Contents

☞ Please, read this document carefully, and **follow the guides strictly**. Failing to do so may result in your project not being evaluated with the consequent loss in marking.

1 About the data challenge

This document provides a description of the data challenge for the Sensing, Perception and Neuroergonomics module of the MRes in Neuroergonomics. The data challenge is a small *hands-on* project that allows the student to implement some of the methods learnt during the course.

Main task: For this data challenge, you will be provided with a database extracted from a blinded real experimental dataset from an fNIRS neuroimaging study on surgical neuroergonomics,. The database was extracted from the original dataset following some common yet basic processing pipeline. The task is to *carry out the last stage of the statistical analysis to confirm or refute some hypothesis* given below.

Bonus task: In addition, you are also provided with the original raw dataset. As a bonus exercise, you are requested to implement your own processing pipeline, and subsequent analysis, and discuss the differences in the findings with both approaches.

Further details of these tasks is provided below in Sect 6.

2 Marking and assessment

The marks obtained for this project are counted towards the final marking of the course. To get the full marks on this data challenge, you are only required to complete the main task of the challenge, that is the statistical analysis from the given database. Notwithstansidng, the bonus exercise can boost your final mark in the challenge if you fall short from perfect marks with the main task. The bonus task is only considered if you complete the main task as well (to passing degree). The data challenge is to be developed outside lecture times. The data challenge ought be developed individually.

See further details in Sect 8.

3 Submission

Upon conclusion of the data challenge, the following evidence has to be produced:

1. A **poster** in `.pdf` format describing the work carried out, the results obtained and conclusions learnt in order to fulfill the data challenge targets (described below).
   This is regardless of whether you only completed the main task or the bonus task as well.

☞ The poster must be submitted via the module BB site. The **deadline for submission is Friday 10th of January 2025 at 16:00**.

The **poster** must be in .pdf format, sized A0 (whether vertical or landscape) and be named:

Surname_Name_SPN_Jan2025.pdf

It is suggested that the content of the poster follows this structure:

1. Title, author name and surname.
2. Abstract (Up to 250 words)
3. Introduction
4. Methods
5. Results.
6. Discussion
7. Conclusions
8. Acknowledgements (Only if required. It may be absent)
9. References.

Ideally, the results must be fully replicable and reproducible from the contents of the poster. It is suggested that the Results section is organised by findings.

4 Goals

The original experiment was conducted to test whether fNIRS neuroimaging is sensitive to surgical expertise related changes at cortical level. However, this goal from the original experiment is only tangential to this data challenge.

Data challenge Goal
To train you on the basics of hypothesis testing and data analytics in the context of fNIRS neuroimaging for sugical neuroergonomics.

Main task: Goal

- Segregational analysis; To tell where the strongest activation is located on each experimental group (intra-group analysis)
- Integrational analysis: To compare the pattern of activation among groups (inter-group analysis)

Bonus task: Goal

- Design a purposeful processing and analysis strategy for studying *segregational* and *integrational* activity from fNIRS experimental data and operationalize it using existing software tools or ad-hoc self developed code.
- To repeat the main task segregational and integrational analysis and estbalish similarities and differences in the findings.

5 Experiment description: Prefrontal Construct Validity

5.1 Experimental Protocol

Details of the original experiment can be found in the original publications [3, 2]. In brief, this is a cross-sectional between subjects single factorial exercise. Each volunteer surgeon performed a surgical reef knot tying task. The exercises were conducted in a classical block design paradigm. The stimulus task was repeated 5 times in self-paced length blocks with interblock rest periods in the following order: 30 seconds initial rest, 5 episodes of self-paced task, followed by 30 seconds rest.

Surgeons of 3 different expertise levels; novices, registrars and consultants participated in the study. Sixty-two, healthy right handed male subjects were recruited for this study; 19 consultant surgeons, 21 surgical trainees, and 22 medical students) You will be able to identify the group from the initial letter of the filename; N for novices, R for registrars and C for consultants.

Experimental task

The experimental task was a reef surgical knot tying task (see companion video in the `data` subfolder). No prior training was provided. The participants remained seated for the duration of the experiment, and were asked to keep their heads as still as possible. During intertrial rest periods, subjects were be asked to keep their eyes closed and hands laying still on a table or on top their knees.

Data acquisition

Functional Near Infrared Spectroscopy (fNIRS) is a biomedical imaging modality capitalizing on light absorption. Infrared light is shined on the scalp and upon exiting it conveys information about tissue oxygenation in the form of relative concentration of oxy- and deoxy-haemoglobin.

For each subject, the functional brain behaviour as proxied from its associated haemodynamics was monitored using a 24-channel continuous wave Near InfraRed Spectroscopy device (NIRS) (Hitachi ETG 4000 OTS) sampled at 10Hz. A configuration of two 3×3 probe arrays was used, composing ten optical fibre sources emitting NIR light (from laser diodes) at 690 and 830 nm and eight optical fibre receivers coupled to avalanche photodiode detectors

Probe Positioning

Probes were positioned according to the UI 10/10 system [1]. See Figure 1 for the approximate probe positioning and channel locations.

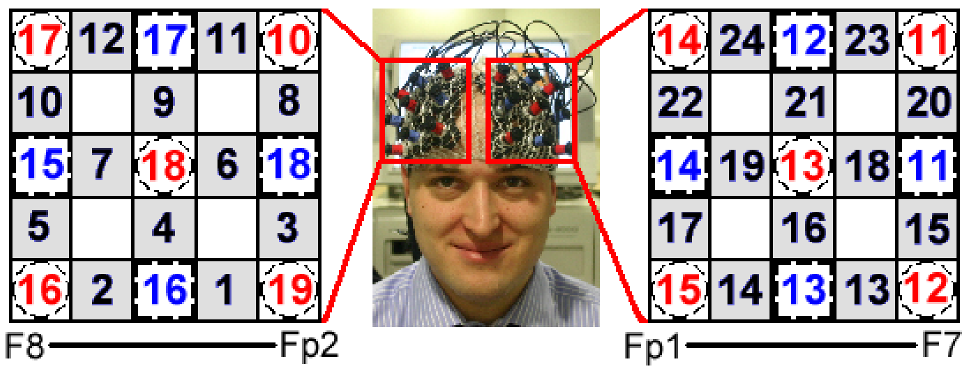

Figure 1: Approximate probe positioning and channel locations.

☞ For this data challenge we shall not be conducting any registration effort. Assume that the channels are correctly located.

6 Tasks description

6.1 Main task

In the `data` subfolder that you are provided, you will find a file name `MainTask/SPNDataChallenge0001.csv`. The data in this database has been extracted for you using a naive processing pipeline using existing software ICNNA, also provided to you in the `code` subfolder as `seriesSPNDataChallenge0001.m`.

The processing pipeline was as follows; optical data was processed using ICNNA [4]. Using the Modified Beer Lambert Law, relative changes in light intensities were converted to changes in haemoglobin (HbO2, HHb and their sum, total haemoglobin HbT), Data was baseline corrected, decimated to 1Hz, and detrended to remove system drift and unrelated physiological signals. Data was averaged across experimental blocks. To overcome temporal variations in knot-tying durations, data was resampled applying a low pass filter. Final datasets comprised 20 baseline values, 37 trial values, and 20 post trial rest values for each NIR channel. Quality checks included checks for saturation relation artefacts, and optode movement.

☞ The provided pipeline described above is indeed very naive, and somewhat short of modern standards. For instance, no bandpass filtering or spatial filtering is applied for a better cleaning of systemic information, optode movement is detected but no correction effort is made (instead data that presents this problem is simply discarded from further processing and analysis), it applies a one-size-fits-all approach to processing, etc. Consequently, although trends for subsequent statistics will be correct, but effect sizes are severely affected and hence interpretation if you are inexperienced may be very challenging. We strongly suggest that you improve over the given pipeline for cleaner results.

Finally, typical statistics including rest and task averages and dispersion, time to peak and nadir, etc were extracted for each subject, channel and Hb species and gethered in the aforementioned database.

As part of this main taks, working from the given data, ***provide statistical evidence confirming or refuting the following hypothesis***:

- Segregational analysis
  - The task induces a lateralized response.
  - In novices, orbital activity is more pronounced than lateral activity.
- Integrational Analysis
  - Prefrontal hemodynamic response as monitored with fNIRS is sensitive to surgical expertise.

You can use any statistical software of your liking. You can apply (or not) any further aggregational operation that you may consider appropriate e.g. block averaging, ROI smoothing, etc, but if you do so, do report that explicitly in your poster.

6.2 Bonus task

For the bonus task, you are provided with a collection of `.csv` files. Each file corresponds to one fNIRS neuroimage. The data is provided raw; this is light measurements data i.e. voltage outputs proportional to light intensity detected by the photodetectors, even before conversion to hemoglobin changes. As aforementioned, data was collected at 2 nominal wavelengths (695 and 830 nm respectively) over 24 channels. The file contains two sections; a header with some metadata and the body of data where each row corresponds to a data sample across all channels.

fNIRS neuroimages are affected by a number of artefacts. The detection and removal of instrumental, physiological and experimental artefacts is a necessary step on any fNIRS processing pipeline before analysis. Unfortunately, there is no a one-size-fits-all processing and analysis strategy that can answer all research questions, plus there is of course the risk to fall into overprocessing.

In this task you will depart from the raw optical neuroimaging data and by designing a purposeful processing and analysis strategy, you will have to show that you are capable of extracting statistically significant segregational and integrational evidence of the differences in the evoked hemodynamics within and between the experimental groups

Methods

As part of this bonus task, you will:

1. Convert this dataset for complaince with `.snirf` and BIDS.
2. Reconstruct the brain hemodynamics as interrogated by fNIRS using the modified Beer-Lambert law with DPF correction.
3. Remove instrumental, physiological and experimental artefacts present in your data.
4. Address data wrangling and data curation aspects to ensure optimal quality of your data.
5. Extract and analyze segregational statistics to show local differences in activation loci.
6. Extract and analyze integrational statistics to show local or global differences in the functional connectivity network.

As a departing point you are provided with a copy of software ICNNA and a minimal script (corresponding to the one used for the main task `seriesSPNDataChallenge0001.m` -see note above regarding some of the limitations of this script-). ICNNA is a software for the analysis of fNIRS neuroimages developed at Imperial College London [4]. However, you are NOT required or enforced to use ICNNA and you are given freedom to use other fNIRS existing packages, and or a combination of them. Other suggested packages are HomER, Brain AnalyzIR and/or MNE-fNIRS.

> ☞ The intellectual property of ICNNA belongs to the Imperial College London. It is given free for academic purpose.

Suggested Analysis

Following the signal reconstruction and processing, for segregational analysis you can rely in the task minus baseline analysis or GLM analysis. Notwithstanding, if you opt for a different analysis, this is also acceptable. Extract typical segregational statistics such as mean task minus baseline signal value, area under the curve, time to peak for HbO2, time to nadir for HbR, etc. Then proceed with inferential statistics as appropriate to the experimental design and observed data distributions.

After you finish the segregational analysis conduct a correlation based integrational analysis. Extract typical graph theory based network statistics such as node degrees, network diameter, clustering coefficients, small world index, etc. And finally, proceed with inferential statistics as appropriate to the experimental design and observed data distributions.

7 Data challenge requirements

- Follow the course project submission rules strictly.
- Data: Provided by the lecturers. Dataset available at the module BB site.
- Solution: The endpoints will be accompanied by the corresponding inferential statistics (reported in APA style).
- Report ALL your endpoints accurately.
- If the bonus task is attempted, follow SfNIRS recommended best practices for reporting [5].

8 Rubric

Your poster submission will be assessed against the following criteria:

- Overall *quality* inc. clarity, organization, conciseness, aesthetics and purposefulness of figures, etc
- *Replicability* of your methods and *reproducibility* of your findings.
- *Correctness* of your statistical approach. If the bonus task is attempted, also about the correctness of your processing pipeline.

> ☞ Favour mathematical rigour rather than “popularity” of a method shall they may be in conflict.

- *Comprehensiveness* of the solution.
- *Critical* and *Creative* capacities. Critical capacity can be demonstrated with a good discussion over you results. Creative capacity can be demonstrated by means of originality and effectiveness in the solution.
- The main task is sufficient to get full marks, but attempting the bonus task can boost your marks substantially.

Bibliography

[1] Valer Jurcak, Daisuke Tsuzuki, and Ippeita Dan. 10/20, 10/10 and 10/5 system revisited: Their validity as a relative head-surface-based positioning systems. *NeuroImage*, 34:1600–1611, 2007.

[2] Daniel Richard Leff, Clare E Elwell, Felipe Orihuela-Espina, Louis Atallah, David T Delpy, Ara W Darzi, and Guang Zhong Yang. Changes in prefrontal cortical behaviour depend upon familiarity on a bimanual co-ordination task: an fnirs study. *Neuroimage*, 39(2):805–813, 2008.

[3] Daniel Richard Leff, Felipe Orihuela-Espina, Louis Atallah, Ara Darzi, and Guang-Zhong Yang. Functional near infrared spectroscopy in novice and expert surgeons–a manifold embedding approach. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 270–277. Springer, 2007.

[4] Felipe Orihuela-Espina, Daniel R. Leff, David R. C. James, Guang-Zhong Yang, and Ara W. Darzi. Imperial college near infrared spectroscopy neuroimaging analysis (ICNNA) framework. *Neurophotonics*, 5(1):011011, 2018.

[5] Meryem A. Yücel, Alexander v. Lühmann, Felix Scholkmann, Judit Gervain, Ippeita Dan, Hasan Ayaz, David Boas, Robert J. Cooper, Joseph Culver, Clare E. Elwell, Adam Eggebrecht, Maria A. Franceschini, Christophe Grova, Fumitaka Homae, Frédéric Lesage, Hellmuth Obrig, Ilias Tachtsidis, Sungho Tak, Yunjie Tong, Alessandro Torricelli, Heidrun Wabnitz, and Martin Wolf. Best practices for fNIRS publications. *Neurophotonics*, 8(01), jan 2021.